

Project #13: Advancing Mobility: Design and Development of a Smart Microprocessor-Controlled Prosthetic Knee

Powered Bionic Lower-Limb Prosthesis with Embedded Control, Adaptive Damping, Tuning Software and Mobile Connectivity



Figure 1. Integrated smart prosthetic knee, embedded electronics, power system, prosthetist tuning software and mobile application.

Project Description and Motivation

This graduation project aims to design, manufacture and validate an affordable smart microprocessor-controlled prosthetic knee for transfemoral (above-knee) amputees. The system combines a load-bearing mechanical knee structure, controllable hydraulic damping, powered actuation, real-time sensing, embedded intelligence and connected software. Its purpose is to provide stable stance support, smooth swing-phase motion and automatic adaptation to the user's speed, terrain and activity while preserving safety, comfort and energy efficiency.

The project supports Egypt's Vision 2030 by promoting inclusive healthcare, local medical-device development and reduced dependence on expensive imported prosthetic technologies. The proposed platform will also provide a reusable research and training system for prosthetists, rehabilitation specialists and mechatronics engineers, enabling future development of intelligent lower-limb prostheses tailored to users in Egypt and the wider region.

Main Objectives

Mechanical platform	Embedded intelligence	Clinical configuration
Develop a compact knee mechanism with controlled resistance, structural strength and safe range of motion.	Fuse sensor data, identify gait phase and command motors/damper in real time.	Provide PC and mobile interfaces for calibration, tuning, monitoring and mode selection.

System Architecture and Hardware Integration

Smart Microprocessor-Controlled Prosthetic Knee

System Architecture for Above-Knee Amputees

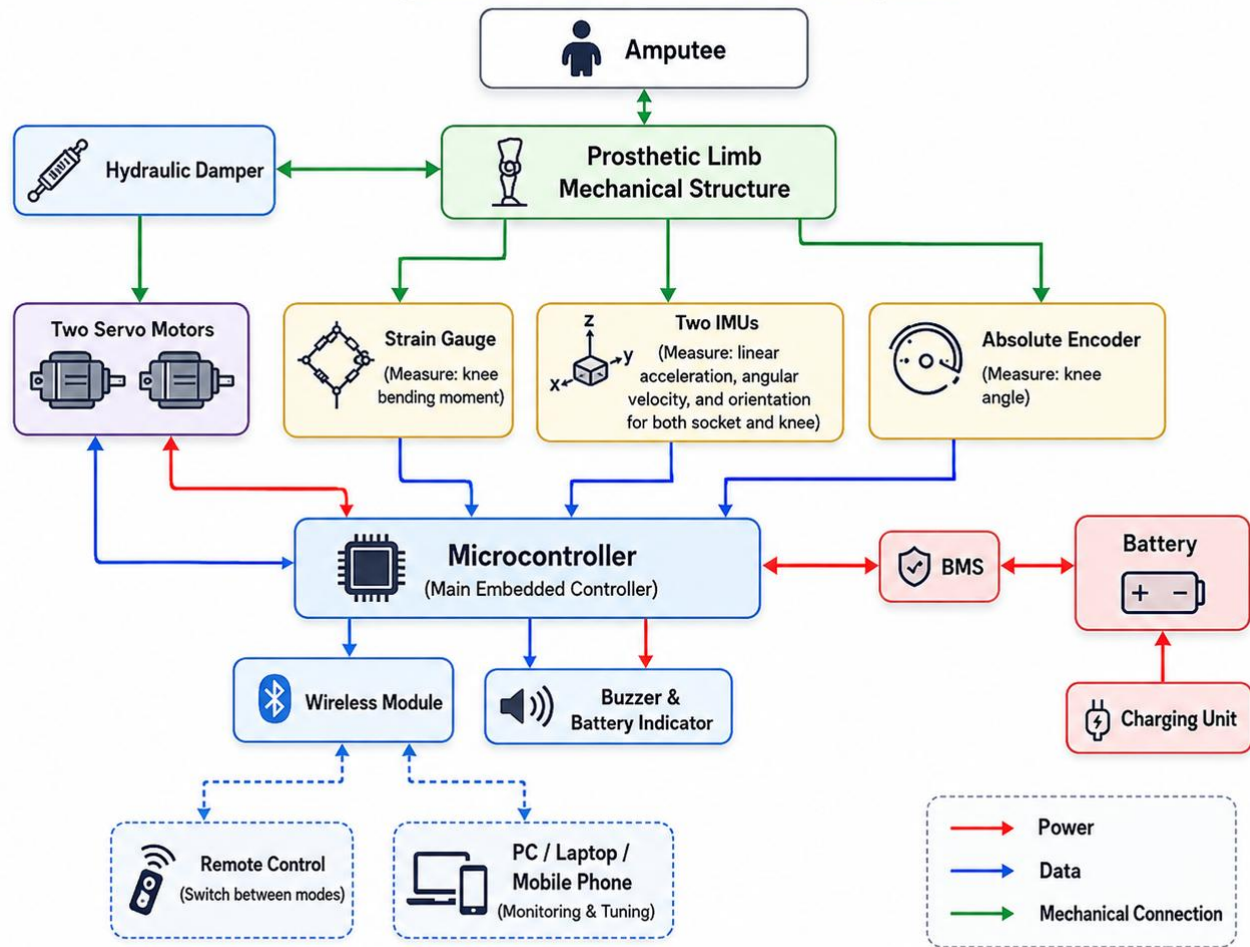


Figure 2. System architecture showing mechanical, sensing, actuation, embedded, power and communication subsystems.

The prosthetic knee is organized as a closed-loop mechatronic system. The mechanical structure transfers the user's load between the socket and prosthetic shank. A hydraulic damper provides controllable resistance for stance stability and swing regulation, while two servo motors can adjust the damping mechanism or provide active assistance for knee flexion and extension. The selected arrangement will be refined through CAD, stress analysis, rapid prototyping and bench testing before integration with the electronic controller.

Key Components and Functions

Sensing and Control	Power, Safety and Communication
Two IMUs on the socket and knee measure acceleration, angular velocity and orientation for gait-state estimation.	A lithium-ion battery pack supplies the motors, controller, sensors and communication electronics.
An absolute encoder measures knee angle precisely; a strain-gauge bridge estimates bending moment and loading.	The BMS monitors voltage, current and temperature, performs protection and estimates state of charge.
The microcontroller performs sensor fusion, gait-phase detection, adaptive control, motor commands, data logging and fault handling.	Bluetooth or another secure wireless link connects the knee to the tuning application, mobile app and mode remote.
Control modes may include stance locking, level walking, stairs, ramps, sit-to-stand and configurable rehabilitation/test modes.	A buzzer, battery indicator, current limits, angle limits and emergency safe state protect the user and hardware.

Embedded Control Strategy

During operation, sensor data are sampled continuously and filtered before being fused into estimates of knee angle, angular velocity, segment orientation, external loading and gait phase. A finite-state or hybrid controller then selects appropriate stance and swing behavior. Resistance, motor torque, damping and safety limits are adjusted according to walking speed, detected activity and personalized parameters. The controller must fail safely: abnormal sensor values, overheating, low battery, overcurrent or excessive angle should trigger warnings and a stable passive configuration.

Software, Tuning, Testing and Expected Outcomes

Smart Microprocessor Prosthetic Knee – End-to-End System Workflow

From sensing to intelligent control, safe power management and connected user experience

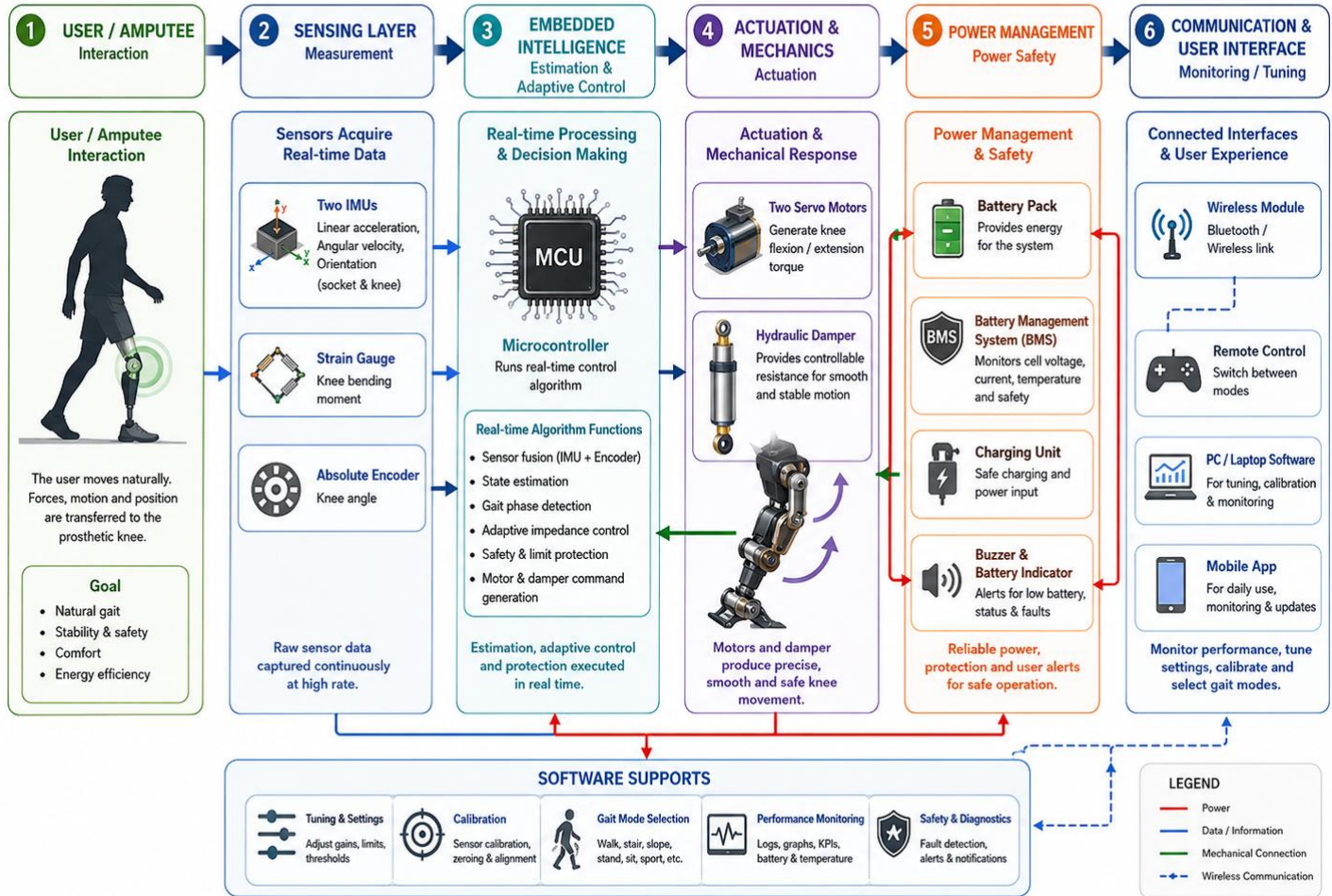


Figure 3. End-to-end workflow from sensing and embedded control to actuation, power management and connected software.

Software and User Interfaces

- Patient profile and activity-level management for individualized configuration.
- Guided sensor calibration, knee alignment checks and zeroing procedures.
- Real-time display of knee angle, IMU signals, estimated load, gait state, motor current, temperature and battery condition.
- Tuning of stance resistance, swing assistance, extension stiffness, damping, thresholds, gains and safety limits.
- Mode selection for level walking, stairs, ramps, standing, sitting and laboratory testing.
- Data logging, event/fault history, firmware updates and export of performance reports.
- A simplified mobile application for status monitoring, battery level and approved user presets.

Development and Evaluation Plan

- Convert the concept into detailed mechanical, electrical and embedded designs with documented interfaces.
- Fabricate and assemble the knee mechanism, sensor mounts, actuator transmission, enclosure and charging system.
- Develop firmware for acquisition, filtering, state estimation, closed-loop control, diagnostics and wireless communication.
- Build the PC tuning application and mobile monitoring interface, then verify communication and cybersecurity basics.
- Perform component, static-load, range-of-motion, cyclic, battery-endurance, thermal and fault-injection tests.
- Use a prosthetic-knee test rig or able-bodied bypass setup before any supervised user evaluation and ethical approval.
- Assess tracking accuracy, response time, gait-phase classification, stability, power consumption, battery runtime and robustness.

Previous Project Videos: <https://www.youtube.com/watch?v=Q5g8UFNkGKk>

| <https://www.youtube.com/watch?v=Gh5KLd81pBs>

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